Phase 9 CVRP Suite Report

Overview

- Condition count: 6.
- Best held-out condition: phase9_closeout_replay_solver_evolution.

Condition Summary

Condition	Mode	Held-out Feasibility	Held-out Penalized Gap	Held-out Feasible Gap	Held-out Runtime (ms)	Accepted Epochs	Fallback Epochs	Generation Errors	Mean Novelty	Mean Complexity
phase9_closeout_baseline_nearest_neighbor_constructive	baseline_nearest_neighbor_constructive	1.0	0.273435	0.273435	42.07024	0	0	0	0.0	0.0
phase9_closeout_baseline_clarke_wright_savings	baseline_clarke_wright_savings	1.0	0.073766	0.073766	74.9565	0	0	0	0.0	0.0
phase9_closeout_baseline_regret_insertion_local_search	baseline_regret_insertion_local_search	1.0	0.466391	0.466391	1518.02508	0	0	0	0.0	0.0
phase9_closeout_direct_generate_plus_one_repair	direct_generate_plus_one_repair	0.0	13.06	n/a	132.75412	1	0	0	0.94602	15.45
phase9_closeout_budget_matched_no_replay	budget_matched_no_replay	0.0	3.0	n/a	214.52828	1	0	0	0.948737	13.93
phase9_closeout_replay_solver_evolution	replay_solver_evolution	1.0	0.065689	0.065689	358.0752	1	0	0	0.884983	15.25

Condition Notes

phase9_closeout_baseline_nearest_neighbor_constructive

- Execution mode: baseline_nearest_neighbor_constructive.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.337704.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.273435.
- Held-out runtime ms: 42.07024.
- Accepted epoch count: 0.
- Fallback epoch count: 0.

- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.002777.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.408032, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.364218, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.317111, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.145642, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.132172, errors=[]

phase9_closeout_baseline_clarke_wright_savings

- Execution mode: baseline_clarke_wright_savings.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.064783.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.073766.
- Held-out runtime ms: 74.9565.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.007791.
- Worst held-out instances:
- X-n247-k50: feasible=True, penalized gap=0.108521, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.099285, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.061249, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.057708, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.042065, errors=[]

phase9_closeout_baseline_regret_insertion_local_search

- Execution mode: baseline_regret_insertion_local_search.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.451049.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.466391.
- Held-out runtime ms: 1518.02508.
- Accepted epoch count: 0.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.086899.
- Worst held-out instances:

- X-n275-k28: feasible=True, penalized gap=0.746105, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.650864, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.395235, errors=[]
- X-n176-k26: feasible=True, penalized gap=0.390341, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.149411, errors=[]

phase9_closeout_direct_generate_plus_one_repair

- Execution mode: direct_generate_plus_one_repair.
- Train feasibility rate: 0.0.
- Train penalized gap: 8.958333.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 13.06.
- Held-out runtime ms: 132.75412.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 1.616667.
- Worst held-out instances:
- X-n275-k28: feasible=False, penalized gap=15.7, errors=["NameError: name 'isinstance' is not defined", 'missing 274 customers']
- X-n247-k50: feasible=False, penalized gap=14.3, errors=["NameError: name 'isinstance' is not defined", 'missing 246 customers']
- X-n223-k34: feasible=False, penalized gap=13.1, errors=["NameError: name 'isinstance' is not defined", 'missing 222 customers']
- X-n190-k8: feasible=False, penalized gap=11.45, errors=["NameError: name 'isinstance' is not defined", 'missing 189 customers']
- X-n176-k26: feasible=False, penalized gap=10.75, errors=["NameError: name 'isinstance' is not defined", 'missing 175 customers']

phase9_closeout_budget_matched_no_replay

- Execution mode: budget_matched_no_replay.
- Train feasibility rate: 0.0.
- Train penalized gap: 3.0.
- Held-out feasibility rate: 0.0.
- Held-out penalized gap: 3.0.
- Held-out runtime ms: 214.52828.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.0.

- Worst held-out instances:
- X-n176-k26: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n190-k8: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n223-k34: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n247-k50: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']
- X-n275-k28: feasible=False, penalized gap=3.0, errors=['solver did not return a list of routes']

phase9_closeout_replay_solver_evolution

- Execution mode: replay_solver_evolution.
- Train feasibility rate: 1.0.
- Train penalized gap: 0.066626.
- Held-out feasibility rate: 1.0.
- Held-out penalized gap: 0.065689.
- Held-out runtime ms: 358.0752.
- Accepted epoch count: 1.
- Fallback epoch count: 0.
- Generation-error epoch count: 0.
- Robustness dispersion across held-out structure families: 0.001023.
- Worst held-out instances:
- X-n176-k26: feasible=True, penalized gap=0.096733, errors=[]
- X-n247-k50: feasible=True, penalized gap=0.077051, errors=[]
- X-n190-k8: feasible=True, penalized gap=0.056596, errors=[]
- X-n275-k28: feasible=True, penalized gap=0.051871, errors=[]
- X-n223-k34: feasible=True, penalized gap=0.046195, errors=[]

Judge Note

Phase-9 CVRP Closeout Suite Summary (Conservative)

Condition	Feasibility (Heldout)	Objective Gap (Heldout)	Runtime (ms)	Robustness Dispersion	Learned vs. Fixed Baseline	Replay vs. Budget-Matched No-Replay
Baseline Nearest Neighbor Constructive	100%	0.2734	42	0.0028	Baseline	N/A
Baseline Clarke-Wright Savings	100%	0.0738	75	0.0078	Baseline	N/A
Baseline Regret Insertion Local Search	100%	0.4664	1518	0.0869	Baseline	N/A
Direct Generate + One Repair	0%	N/A (13.06 penalized)	133	1.617	No (failed feasibility)	N/A
Budget-Matched No Replay	0%	N/A (3.0 penalized)	215	0	No (failed feasibility)	Control for replay

Condition	Feasibility (Heldout)	Objective Gap (Heldout)	Runtime (ms)	Robustness Dispersion	Learned vs. Fixed Baseline	Replay vs. Budget-Matched No-Replay
Replay Solver Evolution	100%	0.0657	358	0.0010	Yes (beat baseline gaps except regret)	Yes (beat budget-matched no replay)

Key Points:

- **Direct synthesis ("direct_generate_plus_one_repair") failed feasibility (0% feasibility), with very high penalized gap and high complexity/novelty.**
- **Fixed baseline constructive heuristics (nearest neighbor, Clarke-Wright, regret insertion local search) all had 100% feasibility. Clarke-Wright had the best gap (0.0738), albeit slower than nearest neighbor. Regret insertion was slowest and worst gap.**
- **Budget-matched independent search without replay had 0% feasibility, penalized gap of 3.0, runtime 215ms, complexity/novelty high, indicating it failed to produce feasible solutions within budget.**
- **Replay-aware iterative search ("replay_solver_evolution") fully feasible (100%), best objective gap (0.0657) better than Clarke-Wright baseline, and reasonable runtime (358 ms). Robustness (dispersion) minimal, complexity similar to direct and budget-matched no replay.**
- **Replay-aware search clearly outperformed fixed baselines in objective gap except regret insertion local search (which was much slower and worse gap).**
- **Replay-aware search also clearly outperformed the budget-matched no-replay control in feasibility, objective gap, and robustness.**

Conclusion:

- The **only learned method that clearly beat fixed baselines was the replay-aware iterative search**.
- **Replay-aware iterative search outperformed the budget-matched independent search without replay.**
- Direct synthesis methods failed to deliver feasible solutions.
- The replay approach showed superior feasibility, better objective gap, competitive runtime, and superior robustness on held-out structure families.

Duration of evaluation: ~8 minutes (464.4 seconds)